Hierarchical Self-Attention Graph Pooling Networks for Leakage Detection of Water Distribution Networks

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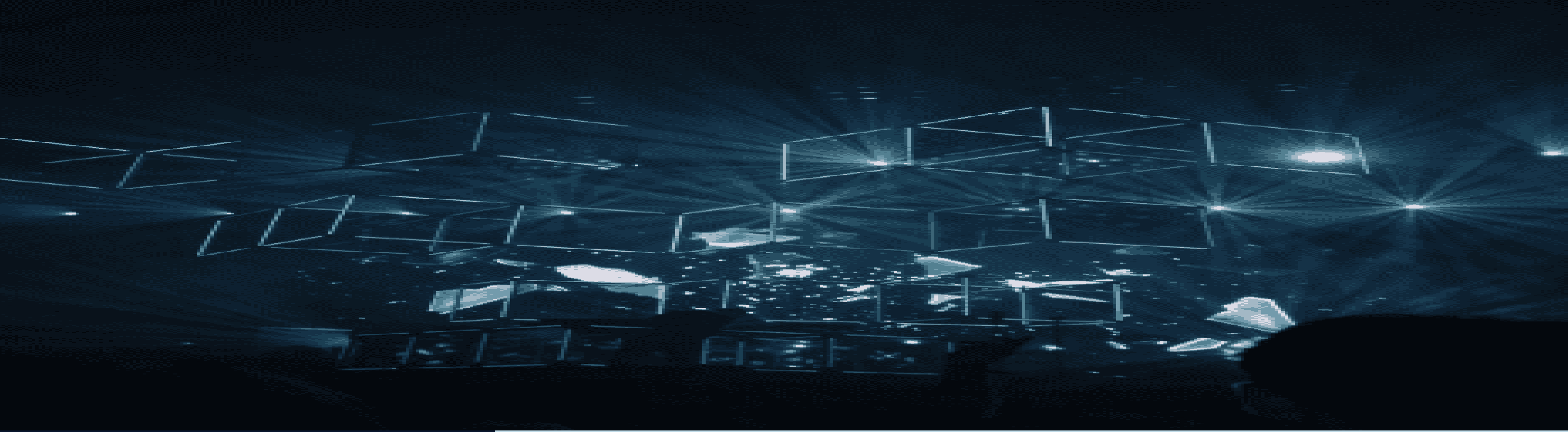


Content

Section 1

Introduction

Importance of Leakage Detection

- **The hazards of pipeline leakage**
  - Wasting water resources
  - Causing energy waste and economic losses
  - Leading to the pollution of drinking water
  - Resulting in secondary disasters

- **National Demands**

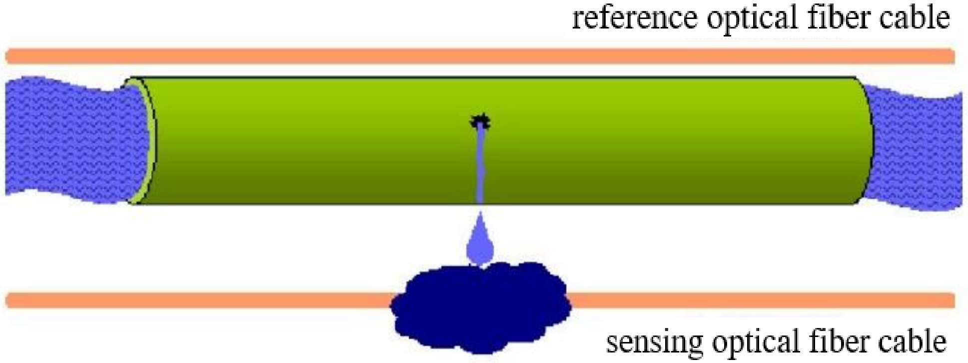
As of 2025: Leakage rate below 9%

Now (before 2020) 84% cities have leakage rate over 9%

The leakage problem is severe

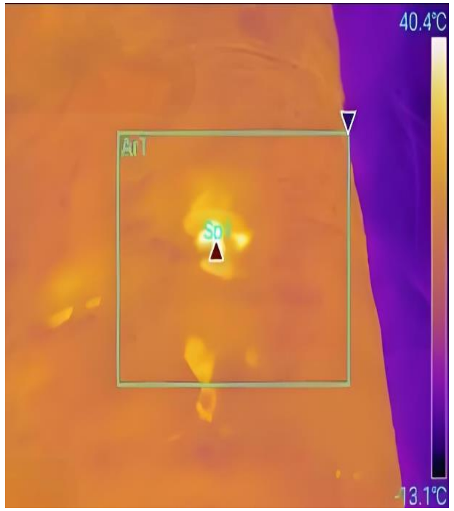
Current Detection Methods

- **Hardware-based detection methods :**
  - Rely on detection equipment
  - Passive detection method
  - Low efficiency and high cost
  - Measurement and analysis of relevant physical parameters of pipelines



distributed optical fiber sensing

listening stick



infrared spectroscopy

ground-penetrating radar

satellite detection

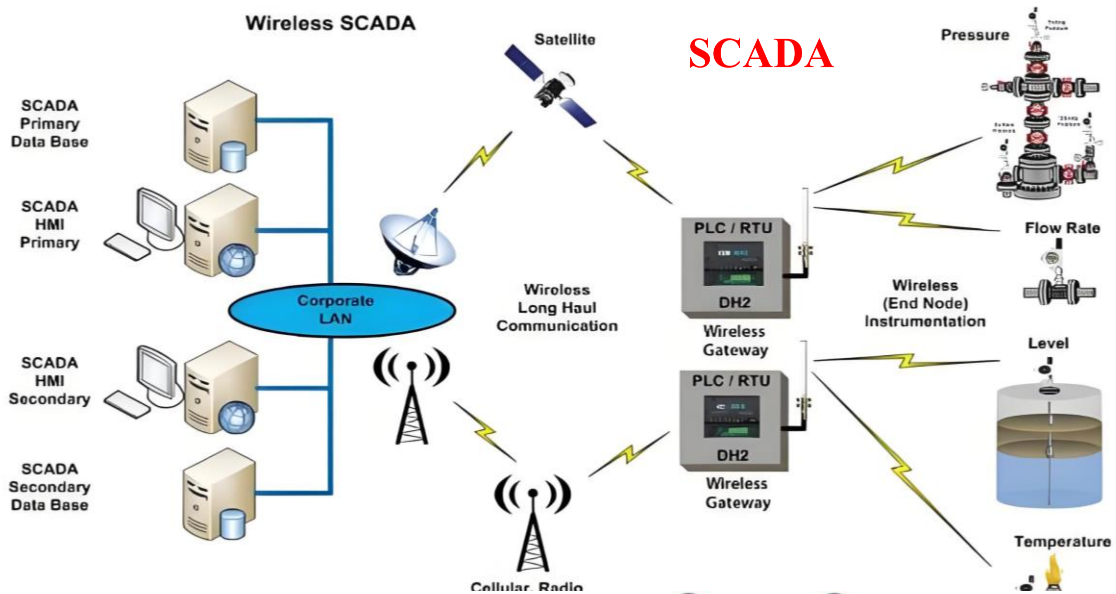
Current Detection Methods

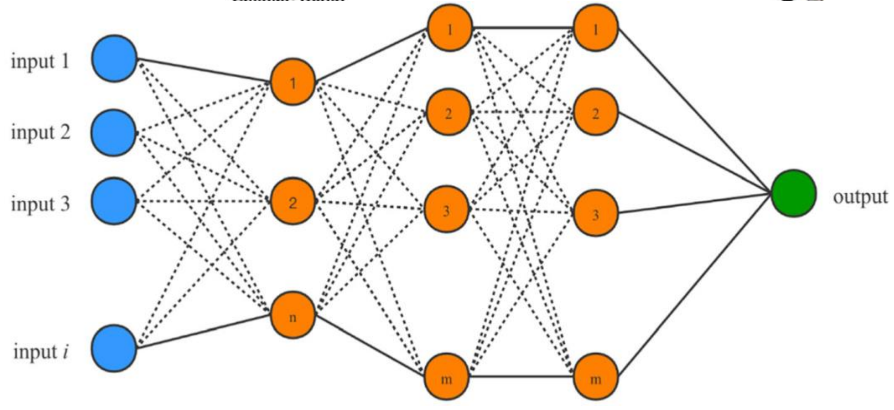
- **Software-based detection methods :**
  - Based on models or algorithms
  - Active detection
  - Real-time data collected from the pipelines
  - Low cost and high efficiency
  - Methods:
    - the balancing approach
    - the Minimum Night Flow (MNF) analysis
    - transient analysis
    - model-based analysis
    - **data-driven approaches:**

      Probability statistics

      Machine learning:Support Vector Machine (SVM) and Artificial Neural Networks (ANN)

      Deep learning: LSTM (Long Short-Term Memory) and CNN

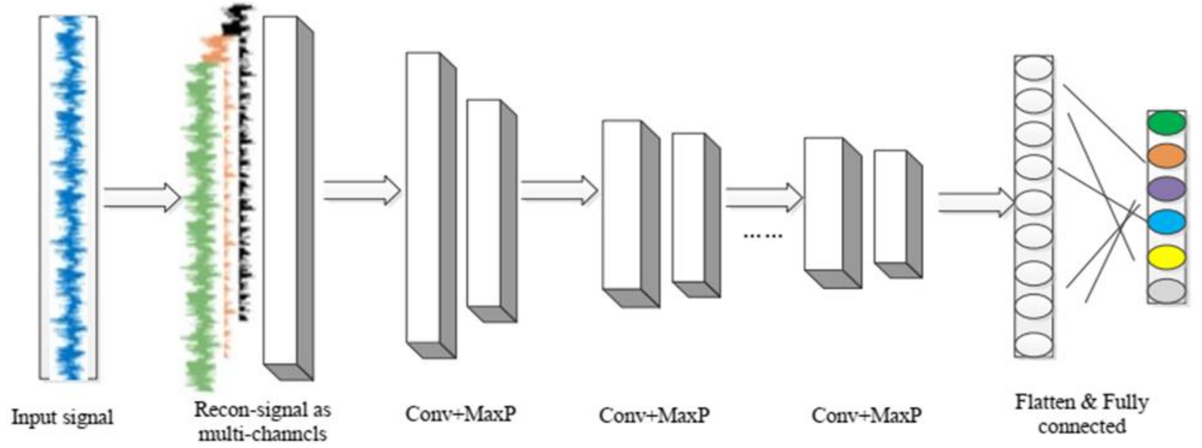




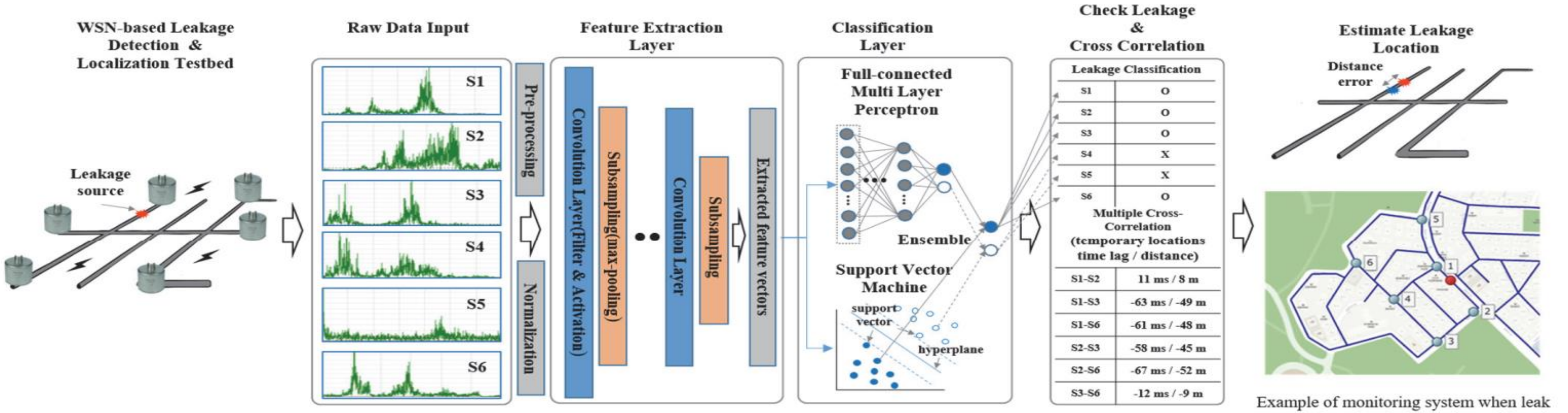
ANN

Traditional deep learning approaches

- **Traditional Methods Limits**:
  - Neglecting the graph relationships between nodes and pipes
  - CNN still has limitations in non-Euclidean structure data
  - **Low accuracy in complex networks**.
  - Few on **large-scale pipe networks**.



CNN

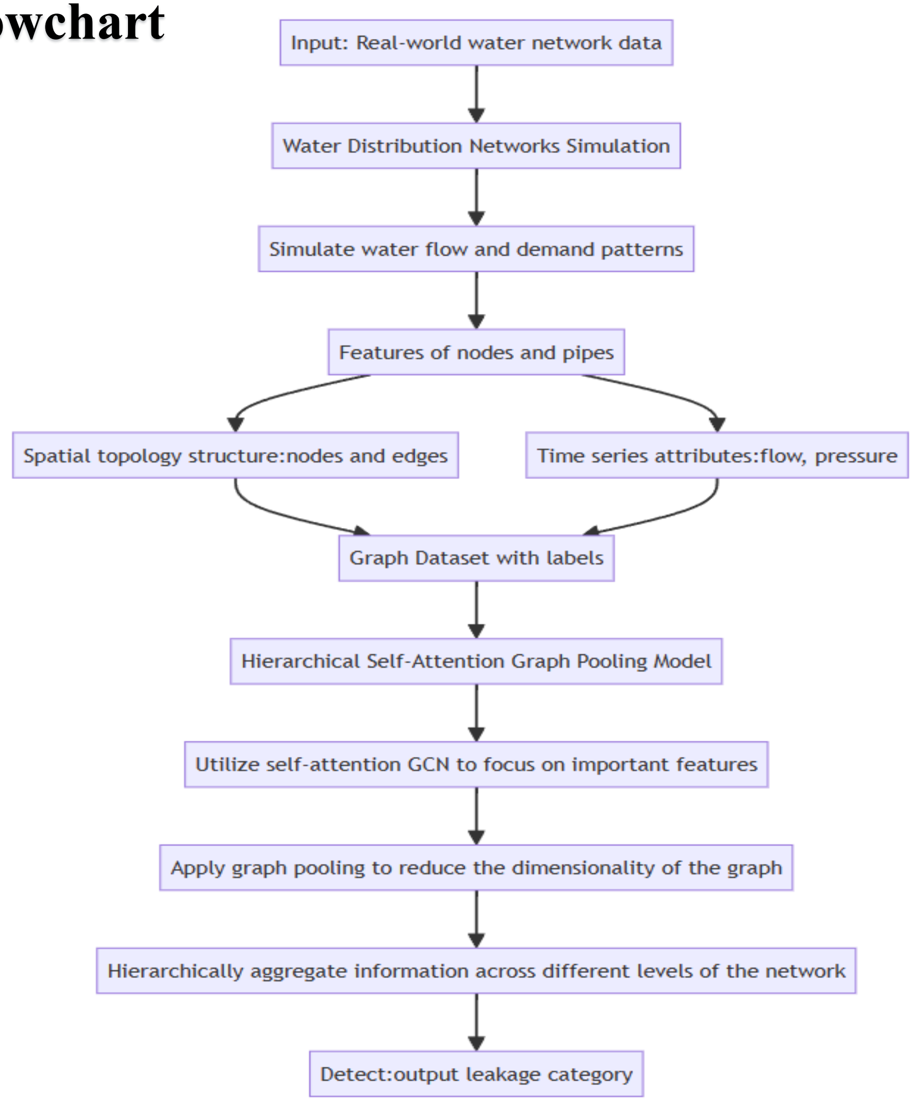


Example of monitoring system when leak occurred, the red point is a estimated location

CNN-SVM

Our Data-driven approaches

Flowchart



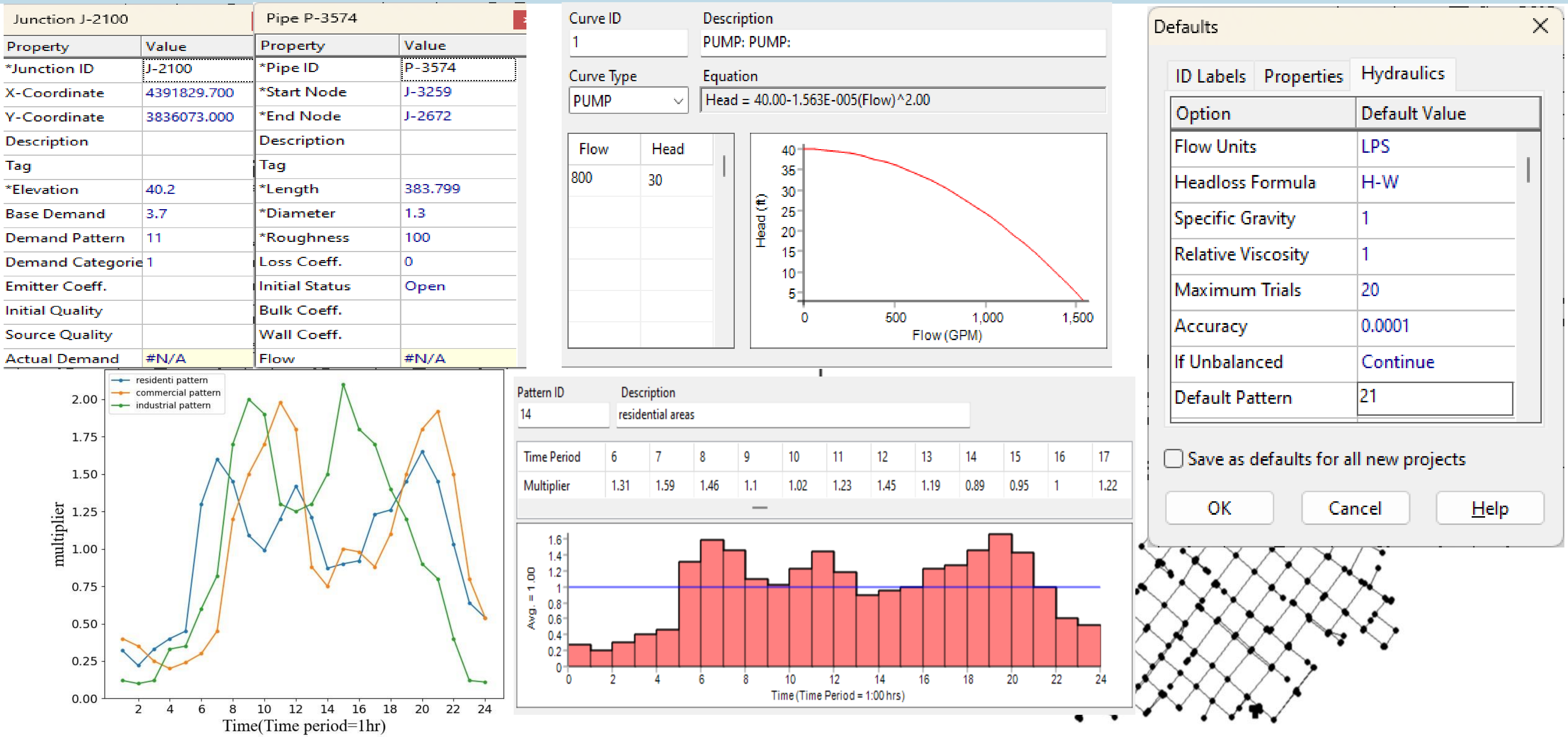
Key points:

- Leakage detection ⟶ Data classification problem
  - Setting leakage nodes, each leaking node labeled with a label
- Dataset ⟶ Graph-structured data
  - Graph-structured with **edges** and **nodes** as basic elements
  - **Adjacency Matrix**
  - Each graph data with a label
- Detect Model ⟶ Hierarchical Self-Attention Graph Pooling Model
  - GCN
  - Self-attention pool
  - Hierarchical pooling structure

Section 2

Simulation of Leakage and Dataset

Water Distribution Network



Data:from the Water Distribution System Research Database.

Leakage Simulation Based On EPANET

➢ Leakage model:

$$Q_i = K_i P_i^n$$

- Q_i —— the leakage flow rate
- K_i —— the emitter coefficient
- P_i —— the pressure at node i
- n —— the pressure exponent (depends on pipe material)

➢ Simulation in EPANET:

Emitter Coeff.	0

- The **emitter coefficient** is an attribute of the node
- The value is directly proportional to the degree of leakage
- 0 —— no leakage

➢ Emitter coefficient:

$$\begin{cases} Q_{ls} = \dfrac{m}{1-m} Q_{sj} \\ C = \dfrac{Q_{ls}}{\left(\dfrac{1}{n}\sum_i^n P_i\right)^{\gamma}} \end{cases}$$

- m —— leakage rate (0<m<1)
- Q_{ls} —— the total leakage
- Q_{sj} —— the actual water demand

➢ Different leakage:

$$C_i = \mu_i C$$

- μ_i —— weight value

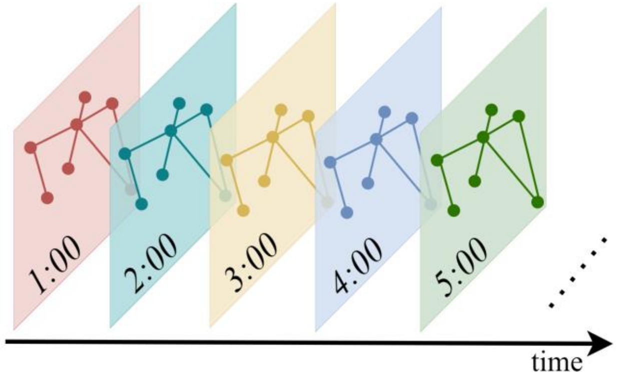
(length, diameter, and material of the pipes)

Dataset

Graph-structured Dataset

- Five nodes serve as leakage points
- Datasets are divided into six distinct categories.
- Continuous 5-day node data

pre-par-ti
wdn_A
wdn_graph_indicator
wdn_graph_labels
wdn_node_features
wdn_node_labels



The graph dataset of time series

The dataset： 668,480 nodes, 708,480 edges, and 720 graphs.

Utilize the simulation step as the data collection interval, each graph is the detection data of all nodes at a specific moment.

(1) wdn_A.txt(720*984行):

sparse (block diagonal) adjacency matrix for all graphs,
each line corresponds to (row, col) resp. (node_id, node_id)

(2) wdn_graph_indicator1.txt(720*859):

column vector of graph identifiers for all nodes of all graphs,
the value in the i-th line is the graph_id of the node with node_id i

(3) wdn_graph_labels.txt (720 lines) :

class labels for all graphs in the dataset,
the value in the i-th line is the class label of the graph with graph_id i

(4) wdn_node_labels.txt (720*859 lines)

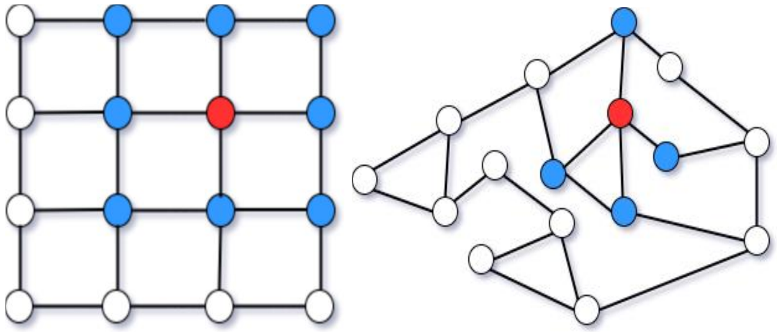
column vector of node labels,
the value in the i-th line corresponds to the node with node_id i

Section 3

Hierarchical Self-Attention Graph Pooling Model

Attention ——achieved by GCN

Graph Convolutional



2-D convolution and graph convolution

$$X_{\ell+1} = \sigma(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}X_\ell W_\ell)$$

σ —— activation function(*tanh*)

$\hat{A}$ —— *adjacency matrix with self-connections*

$\hat{A} = A + I$

$\hat{D}$ —— *degree matrix*

$L\text{sym} = \hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}$

Attention Score

general structure:

$$Z = \sigma(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}X\Theta_{att})$$

Variations

edge augmentation:

$$Z = \sigma(\mathrm{GNN}(X, A + A^2))$$

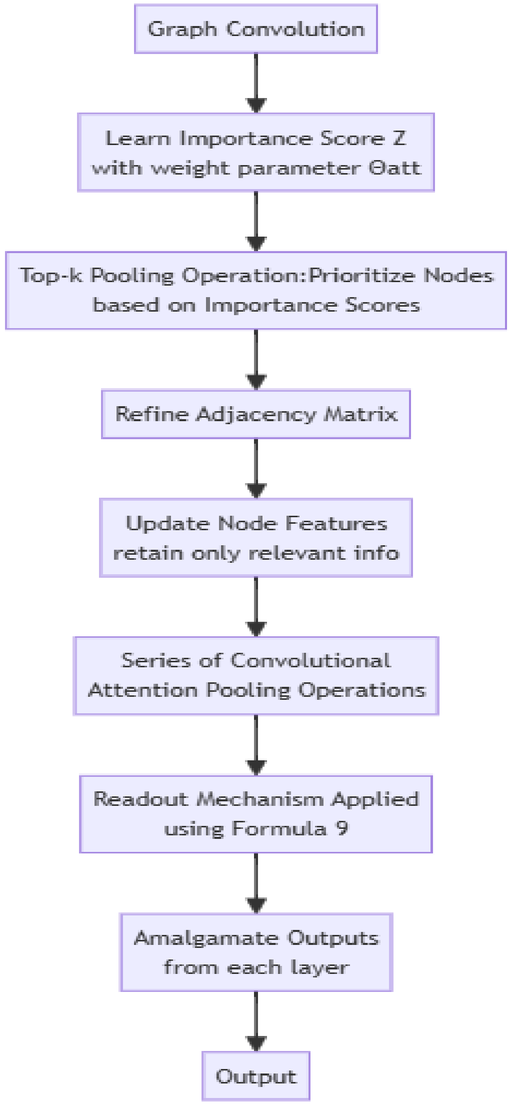
stack of GNN layers:

$$Z = \sigma(\mathrm{GNN}_2(\sigma(\mathrm{GNN}_1(X, A)), A))$$

multiple GNNs:

$$Z = \frac{1}{M}\sum_m \sigma(\mathrm{GNN}_m(X, A))$$

Hierarchical pooling structure



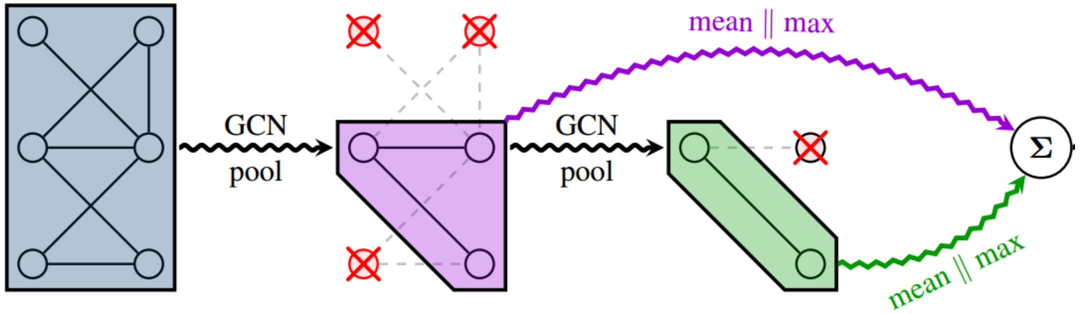
readout mechanism

- a mean pooling and maximum pooling

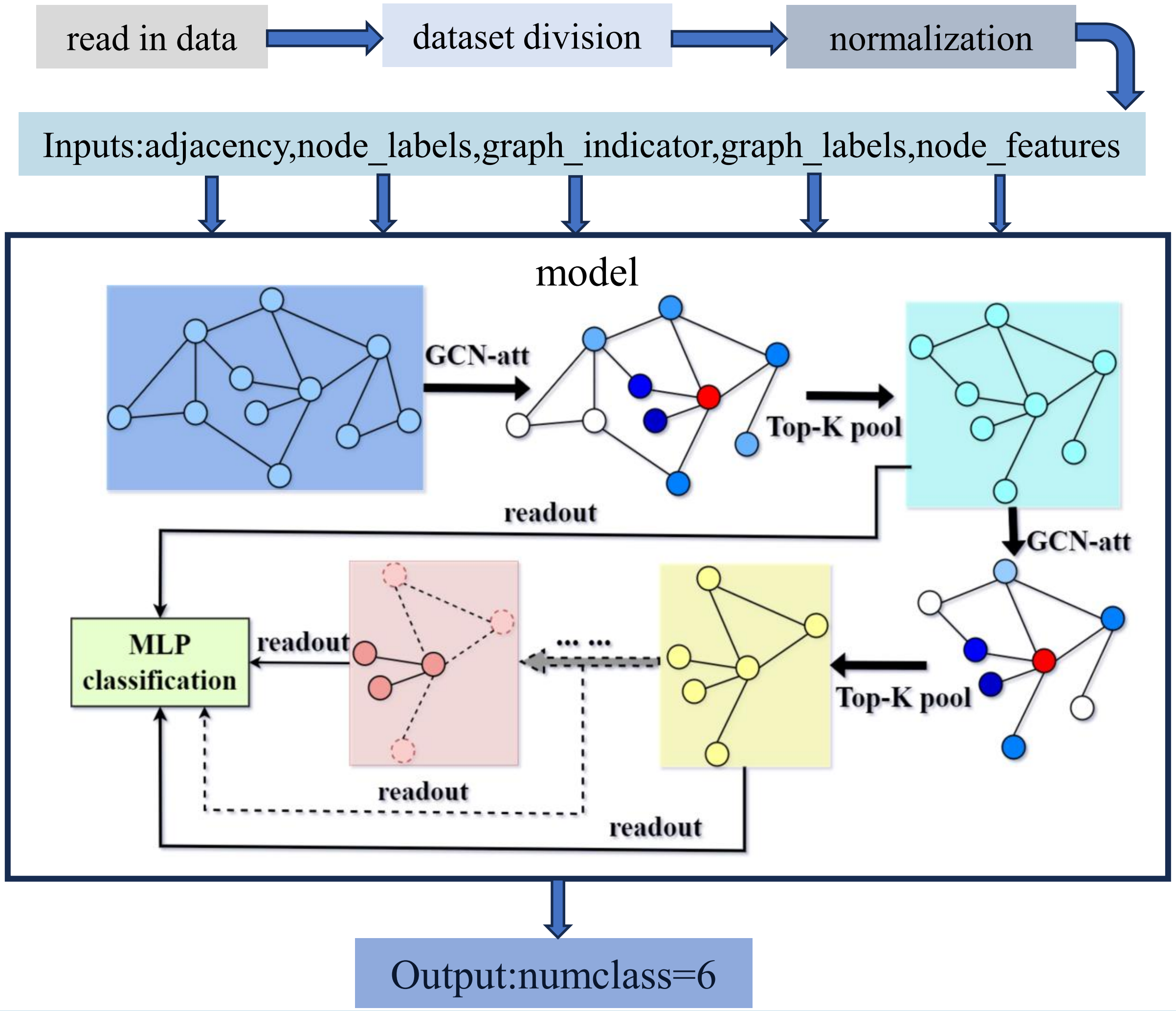
$$s = \frac{1}{N}\sum_{i=1}^{N} x_i \,\|\, \max_{i=1}^{N} x_i$$

- concatenation

$$\vec{s} = \sum_{l=1}^{L} \vec{s}^{(l)}$$



Hierarchical Self-Attention Graph Pooling Model



➢ Details

- Graph Convolutional Pooled Blocks —— Four
- GCN:
  input —— node_features
  input_dim = 4
- Self-attention pooling layers:
  retention rate(0.3*2 layers,0.5*2layers)
  output_dim = 1
- Four readout layers:
  extract and aggregate feature
  hybrid of maximum and average pooling
- MLP for classification
  output_dim=6
  predicted_classes = output.argmax(dim=1)

Section 4

Experiments and Analysis

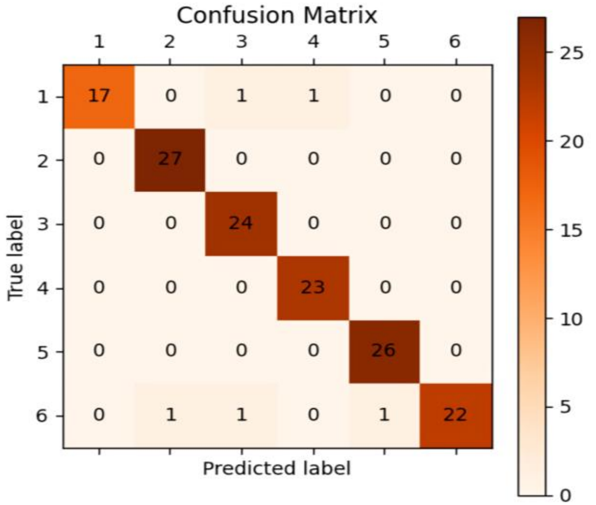
Experimental Results

Experiment configuration

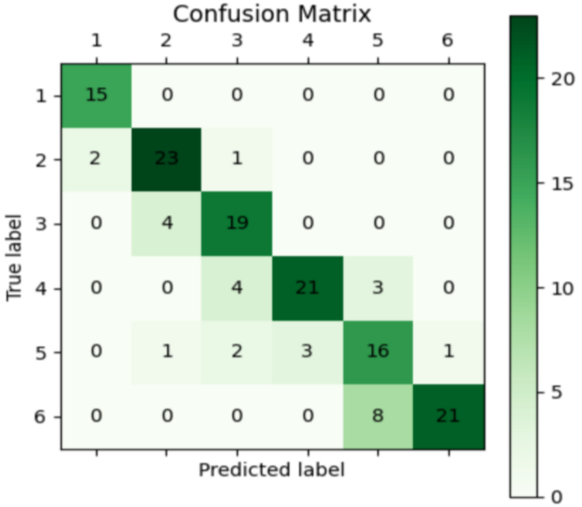
- Operating system: Windows
- Hardware requirements:
  - -NVIDIA GPU
  - -CUDA 11.3
  - -CuDNN 8.0.5
- Python version: 3.9
- Parameters:
  - -dataset division 8:1:1
  - -Adam
  - - Crossentropy loss
  - -initial lr:0.005
    lr_scheduler
  - -Epoch:3100

Leakage label	Precision	Recall	F1-score	Support
1	1.00	0.89	0.94	19
2	0.96	1.00	0.98	27
3	0.92	1.00	0.96	24
4	0.96	1.00	0.98	23
5	0.96	1.00	0.98	26
6	1.00	0.88	0.94	25

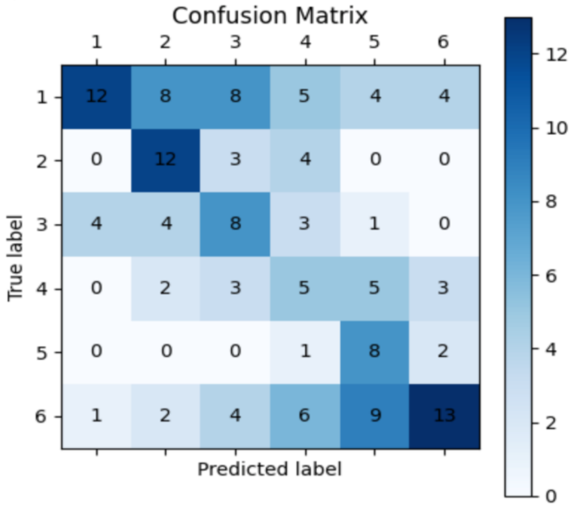
Accuracy rate of 92% within the test dataset



hierarchical self-attention pool model



global self-attention pool model



GCN-only model

Thank You